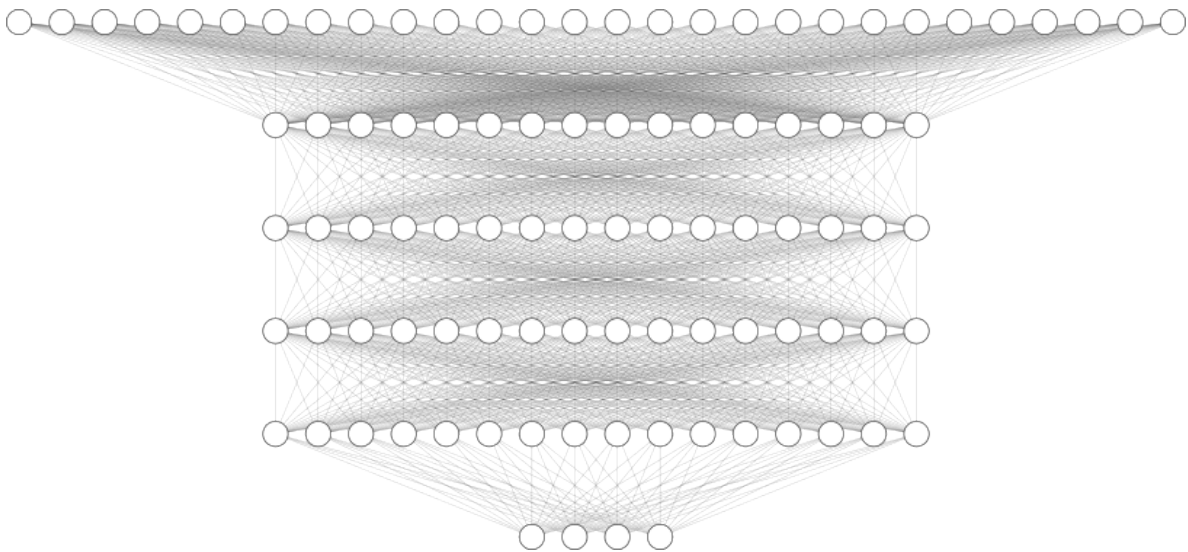


PPGEE2249 Aprendizado de Máquina
Assignment 3
Prof. Daniel Guerreiro e Silva

Thiago Tomás de Paula
November 30, 2025



► Question 1

Train and evaluate a Feedforward Neural Network (FNN) for a multiclass classification task. Split your data into a training set, a validation set (used to define the stopping criterion - e.g., number of epochs), and a test set for the final evaluation of the model. The output layer should use one Softmax neuron per class, and the training objective should be the minimization of cross-entropy loss. Describe your dataset, the network architecture (hidden units, layer dimensions, and number of layers), and discuss the overall performance (data splitting strategy, classification accuracy, and confusion matrix).

► Question 2

Train and evaluate a Decision Tree (DT) on the same classification problem chosen in Question 1. Present the resulting tree (depth, rule-based description) and compare its performance with the results obtained in Question 1.

Dataset description. The dataset chosen is Michael J. Fox Foundation’s Parkinson’s Progression Markers Initiative (PPMI) bank of raw sequencing reads from blood samples. Over 4000 participants have had their RNA transcribed across several visits, and the result of this analysis bundled with other categorical variables (such as age, gender, race, etc.) to form the PPMI dataset. As a dataset designed for the study of Parkinson Disease (PD) prediction and identification, each patient falls under:

- “0”/“Case” if with PD
- “1”/“Control” if healthy
- “2”/“Other” if prodromal

Of the original 56 columns in the CSV of PPMI patient metadata (`meta_data11192021.csv`)¹, 4 are diagnostic labels, and 23 were dropped from consideration due to one or more of the following:

- (i) redundancy (e.g., “Age (Bin)”, “Age at diagnosis”);
- (ii) could not contribute to the cause of PD (e.g., “Month” of visit, “Plate” of the sample);
- (iii) are patient identifiers (e.g., “HudAlphaSampleName”, “PATNO”);
- (iv) were sparsely present across samples (e.g., “UPSIT”).

As such, each sample from the dataset consists of 29 features. From this subset of data, features with correlation greater than or equal to 90% (in absolute value) were dropped.

After the removal of highly-correlated features, the samples are shuffled and finally separated into training, validation, and test sets in splits of 50%, 25% and 25%, respectively.

Table 1 profiles the sets obtained from the preprocessing of PPMI. Figure 1 shows a flowchart of the preprocessing done to data. Figure 2 shows the correlation matrix of the PPMI dataset. Since one “Neutrophils (%)” was dropped due to high negative correlation with “Lymphocytes (%)”, there are 28 features total instead of 29.

Table 1: PPMI splits. Randomizer seed was fixed at 2421.

Class	Absolute and relative amounts within split					
	Train		Validation		Test	
0 (PD)	183	30.30%	100	33.11%	112	36.96%
1 (Healthy)	416	68.87%	200	66.23%	189	62.38%
2 (Prodromal)	5	0.83%	2	0.66%	2	0.66%
Samples	604	49.96%	302	24.98%	303	25.06%

¹See the note on data at the end of this report for a more detailed explanation on the features.

Figure 1: Steps taken in this work to prepare data for machine learning algorithms.

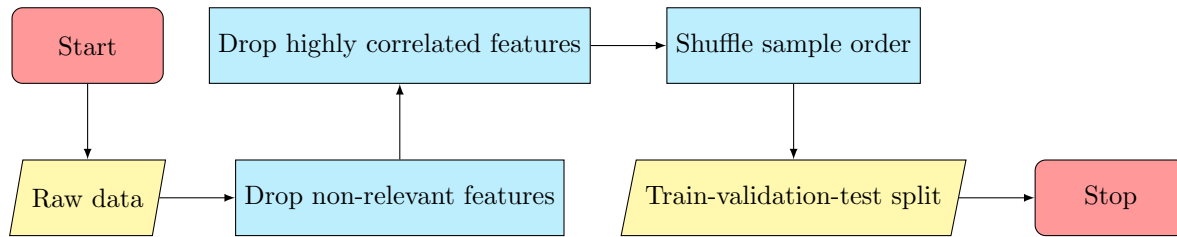


Figure 2: Feature correlations found in a subset of PPMI.

Genetic Status	1.00	-0.05	0.01	-0.07	0.04	-0.05	0.00	-0.01	-0.03	-0.04	0.04	-0.04	-0.02	0.01	-0.05	0.02	-0.11	-0.01	-0.09	0.05	0.01	-0.04	-0.02	-0.05	0.04	0.05	0.04	0.01
Sex	-0.05	1.00	-0.03	-0.09	0.07	0.06	-0.01	0.02	-0.01	0.10	-0.14	0.13	0.02	-0.57	0.84	-0.01	0.08	0.01	0.12	-0.04	0.05	0.02	-0.01	-0.04	-0.02	0.12	0.09	-0.15
Ethnicity	-0.01	-0.03	1.00	0.12	-0.05	-0.05	0.00	-0.01	-0.03	-0.04	0.03	-0.04	-0.03	0.02	-0.03	-0.03	-0.03	0.03	-0.03	0.04	0.07	-0.03	-0.01	0.04	-0.03	-0.01	0.03	-0.09
Race	-0.07	-0.09	0.12	1.00	-0.03	-0.04	0.06	-0.05	-0.05	0.01	0.09	-0.04	0.10	0.01	-0.11	-0.08	0.04	0.06	-0.01	0.08	-0.01	0.11	-0.00	-0.01	0.02	-0.00	0.01	-0.13
Education Level	-0.04	0.07	-0.05	-0.03	1.00	0.15	0.00	-0.02	-0.04	0.04	0.02	-0.02	-0.04	0.04	0.11	0.06	0.00	-0.06	0.07	-0.01	-0.01	-0.05	-0.05	0.04	0.00	0.02	0.00	-0.04
Age at Consent	-0.05	0.06	-0.05	-0.04	0.15	1.00	0.01	0.11	-0.04	0.05	-0.14	0.15	0.00	0.01	0.08	0.04	-0.05	-0.06	0.02	-0.15	0.12	-0.01	-0.03	-0.00	0.13	0.10	0.14	-0.33
RIN Value	-0.00	-0.01	0.00	0.06	0.00	0.01	1.00	-0.02	-0.00	0.01	-0.02	0.02	0.28	-0.05	-0.04	0.04	0.00	0.01	0.08	0.05	0.11	0.30	0.00	-0.01	-0.02	-0.03	-0.07	0.06
Neutrophil Score	-0.01	0.02	-0.01	-0.05	-0.02	0.11	-0.02	1.00	-0.17	-0.19	-0.84	0.81	-0.09	0.21	0.14	-0.07	-0.12	-0.32	0.18	-0.62	0.55	0.07	-0.04	-0.01	0.18	0.19	0.21	-0.11
Basophils (%)	-0.03	-0.01	-0.03	-0.05	-0.04	-0.04	-0.00	-0.17	1.00	0.18	0.09	-0.14	-0.01	-0.02	-0.02	0.02	0.03	0.03	-0.01	0.06	-0.12	-0.02	0.02	0.01	-0.01	-0.03	-0.04	0.02
Eosinophils (%)	-0.04	0.10	-0.04	0.01	0.04	0.05	0.01	-0.19	0.18	1.00	0.03	-0.11	0.02	-0.08	0.06	-0.02	0.06	-0.04	0.08	0.07	-0.13	0.02	0.06	0.03	-0.05	-0.03	-0.04	-0.05
Lymphocytes (%)	-0.04	-0.14	0.03	0.09	0.02	-0.14	-0.02	-0.84	0.09	0.03	1.00	-0.89	0.04	-0.01	-0.17	0.09	0.09	0.37	-0.21	0.55	-0.52	-0.07	0.05	0.04	-0.16	-0.17	-0.18	0.14
Neutrophil/Lymphocyte	-0.04	0.13	-0.04	-0.04	-0.02	0.15	0.02	0.81	-0.14	-0.11	-0.89	1.00	-0.05	0.02	0.15	-0.06	-0.10	-0.35	0.20	-0.52	0.51	0.05	-0.03	-0.03	0.21	0.20	0.21	-0.16
PC1	-0.02	0.02	-0.03	0.10	-0.04	0.00	0.28	-0.09	-0.01	0.02	0.04	-0.05	1.00	-0.28	-0.15	-0.18	-0.12	0.17	0.21	-0.06	-0.16	0.84	0.03	-0.08	0.03	0.00	-0.04	0.00
PC2	-0.01	-0.57	0.02	0.01	0.04	0.01	-0.05	0.21	-0.02	-0.08	-0.01	0.02	-0.28	1.00	-0.05	0.10	0.09	-0.08	-0.10	0.01	-0.05	-0.17	-0.03	0.09	0.03	-0.06	-0.01	0.08
PC3	-0.05	0.84	-0.03	-0.11	0.11	0.08	-0.04	0.14	-0.02	0.06	-0.17	0.15	-0.15	-0.05	1.00	0.06	0.11	-0.02	0.04	-0.03	0.02	-0.09	-0.04	-0.01	-0.00	0.10	0.09	-0.12
PC4	-0.02	-0.01	-0.03	-0.08	0.06	0.04	0.04	-0.07	0.02	-0.02	0.09	-0.06	-0.18	0.10	0.06	1.00	0.17	0.03	-0.11	-0.14	-0.06	-0.47	-0.12	0.14	0.05	0.07	0.04	0.03
PC5	-0.11	0.08	-0.03	0.04	0.00	-0.05	0.00	-0.12	0.03	0.06	0.09	-0.10	-0.12	0.09	0.11	0.17	1.00	0.09	-0.05	-0.07	-0.11	-0.06	0.06	0.13	-0.08	-0.09	-0.08	0.01
PC6	-0.01	0.01	0.03	0.06	-0.06	-0.06	0.01	-0.32	0.03	-0.04	0.37	-0.35	0.17	-0.08	-0.02	0.03	0.09	1.00	-0.31	-0.05	0.03	0.02	0.00	0.23	-0.07	-0.05	-0.11	0.02
PC7	-0.09	0.12	-0.03	-0.01	0.07	0.02	0.08	0.18	-0.01	0.08	-0.21	0.20	0.21	-0.10	0.04	-0.11	-0.05	-0.31	1.00	-0.19	-0.09	0.27	0.07	0.34	0.10	0.11	0.13	-0.10
PC8	-0.05	-0.04	0.04	0.08	-0.01	-0.15	0.05	-0.62	0.06	0.07	0.55	-0.52	-0.06	0.01	-0.03	-0.14	-0.07	-0.05	-0.19	1.00	-0.21	-0.09	0.06	0.03	-0.14	-0.17	-0.15	0.12
PC9	-0.01	0.05	0.07	-0.01	-0.01	0.12	0.11	0.55	-0.12	-0.13	-0.52	0.51	-0.16	-0.05	0.02	-0.06	-0.11	0.03	-0.09	-0.21	1.00	-0.04	-0.05	0.08	0.10	0.13	0.12	-0.10
mRNA (%)	-0.04	0.02	-0.03	0.11	-0.05	-0.01	0.30	0.07	-0.02	0.02	-0.07	0.05	0.84	-0.17	-0.09	-0.47	-0.06	0.02	0.27	-0.09	-0.04	1.00	0.06	-0.20	0.03	-0.00	-0.03	-0.00
globin (%)	-0.02	-0.01	-0.01	-0.00	-0.05	-0.03	0.00	-0.04	0.02	0.06	0.05	-0.03	0.03	-0.03	-0.04	-0.12	0.06	0.00	0.07	0.06	-0.05	0.06	1.00	0.37	-0.03	0.01	0.02	0.01
rRNA (%)	-0.05	-0.04	0.04	-0.01	0.04	-0.00	-0.01	-0.01	0.01	0.03	0.04	-0.03	-0.08	0.09	-0.01	0.14	0.13	0.23	0.34	0.03	0.08	-0.20	0.37	1.00	0.07	0.08	0.01	-0.05
UPDRS1 score	-0.04	-0.02	-0.03	0.02	0.00	0.13	-0.02	0.18	-0.01	-0.05	-0.16	0.21	0.03	0.03	-0.00	0.05	-0.08	-0.07	0.10	-0.14	0.10	0.03	-0.03	0.07	1.00	0.64	0.43	-0.25
UPDRS2 score	-0.05	0.12	-0.01	-0.00	0.02	0.10	-0.03	0.19	-0.03	-0.03	-0.17	0.20	0.00	-0.06	0.10	0.07	-0.09	-0.05	0.11	-0.17	0.13	-0.00	0.01	0.08	0.64	1.00	0.69	-0.23
UPDRS3 score	-0.04	0.09	0.03	0.01	0.00	0.14	-0.07	0.21	-0.04	-0.04	-0.18	0.21	-0.04	-0.01	0.09	0.04	-0.08	-0.11	0.13	-0.15	0.12	-0.03	0.02	0.01	0.43	0.69	1.00	-0.26
moca	-0.01	-0.15	-0.09	-0.13	-0.04	-0.33	0.06	-0.11	0.02	-0.05	0.14	-0.16	0.00	0.08	-0.12	0.03	0.01	0.02	-0.10	0.12	-0.10	-0.00	0.01	-0.05	-0.25	-0.23	-0.26	1.00
Genetic Status																												
Sex																												
Ethnicity																												
Race																												
Education Level																												
Age at Consent																												
RIN Value																												
Neutrophil Score																												
Basophils (%)																												
Eosinophils (%)																												
Lymphocytes (%)																												
Neutrophil/Lymphocyte																												
PC1																												
PC2																												
PC3																												
PC4																												
PC5																												
PC6																												
PC7																												
PC8																												
PC9																												
mRNA (%)																												
globin (%)																												
rRNA (%)																												
UPDRS1 score																												
UPDRS2 score																												
UPDRS3 score																												
moca																												

Visit <https://www.ppmi-info.org/access-data-specimens/data> for more details on PPMI. Note that download of the metadata CSV requires applying for the data access.

FNN description. The FNN structure was obtained by minimizing validation error rate within a small search space of hyperparameter values. The hyperparameters considered and the associated search space boundaries are shown in table 2's middle column.

Table 2: FNN hyperparameters.

Hyperparameter	Search space	Best after 100 trials
Hidden layers' activation function	ReLU* or sigmoid	ReLU
Hidden layers' width	$\{1, 2, \dots, 16\}$	16
Number of hidden layers	$\{1, 2, \dots, 16\}$	4
Learning rate**	$[10^{-4}, 10^{-2}]$	1.88×10^{-3}

* Rectified Linear Unit, $f(x) = \max(0, x)$

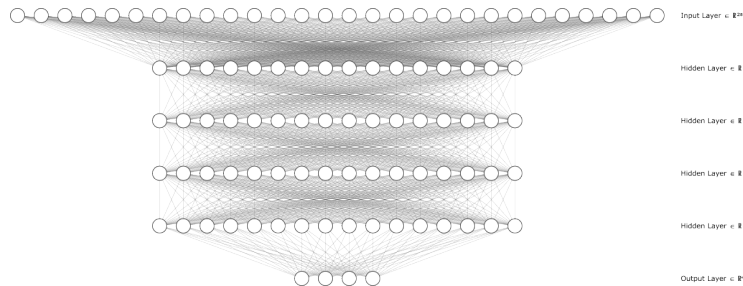
** Sampled from the log domain

More precisely, the validation error corresponds to the number of wrong predictions made on the validation samples divided by the number of validation samples.

To find the point of minimum validation error within the hyperparameter space, the **Optuna framework** was used. This tool picks a subspace of the previously defined search space, samples hyperparameters from it, and evaluates the model using those values. The exploration of the subspace as a whole is called trial, and the model evaluations within a trial, steps. Trials are repeated arbitrarily, and their results are saved to guide following searches, with unpromising trials were stopped (pruned) early.

The FNN with smallest validation error, 4.30%, has the parameters presented in table 2's right column. Figure 3 illustrates this network. Trials used 15 epochs, 32 batch size, and the Adam optimizer from the Keras Python library for gradient descent calculation. No dropout or batch normalization was applied due to the relatively small size of the neural network.

Figure 3: FNN with smallest validation error after 100 trials.

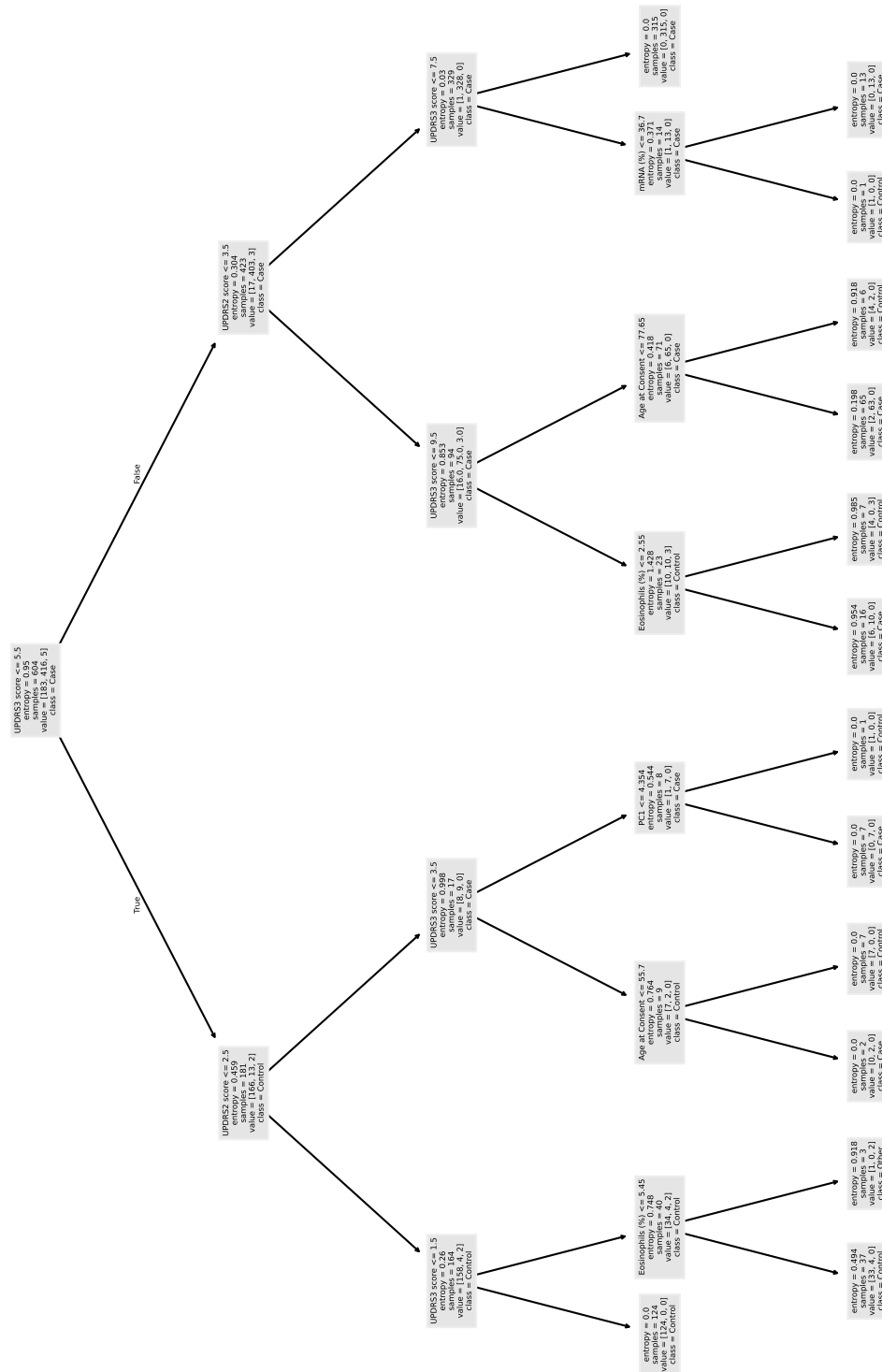


DT description. The DT's structure was obtained in the same fashion as the FNN's, that is, by minimizing validation error on a hyperparameter space. The hyperparameters considered and the associated search space boundaries are shown in table 3. The best DT with smallest validation error, 4.97%, has the parameters presented in table 3. Figure 4 illustrates this network. .

Table 3: DT hyperparameters.

Hyperparameter	Search space	Best after 100 trials
Criterion	Entropy, Gini, or Log-loss	Entropy
Maximum depth	$\{2, 3, \dots, 20\}$	4

Figure 4: DT with smallest validation error after 100 trials.



Performance comparison.

The low error rates during validation strongly suggest the FNN has compatible complexity with respect to the multiclassification task and hand, and that the DT is capable of using only the more impactful features - such as UPDRS scores and age - when diagnosing for PD. These models were ran on the test split, resulting in the confusion matrices shown in fig. 5 and metrics listed in table 4. It is fair to affirm that both models had similarly great performance on classfying both healthiness and PD, although DT has slightly fewer misclassifications in these cases.

With that said, both models showed very poor results with respect to the prodromal classifications, with 0% accuracy. This is, partly, an effect of the optimization objective set during validation: since models were being selected on overall error rate, and not classwise error rates, the frequency of classes in the split becomes an influential factor on performance. In other words, the models tolerate error on prodromal classifications because the number of prodromal patients is small when compared to healthy and PD cases.

Future expansions on this study would consider applying or developing a way to make prodromal samples more influential during validation. One way to do this is making the objective function multivariate, e.g., minimizing not the overall classification error rate, but a vector of classwise error rates: since the error rate on classes with a small amount of samples is sensitive to misclassifications, the proposed objective function would favor it over the error rate of other classes.

Figure 5: Confusion matrices of best FNN and DT architectures.

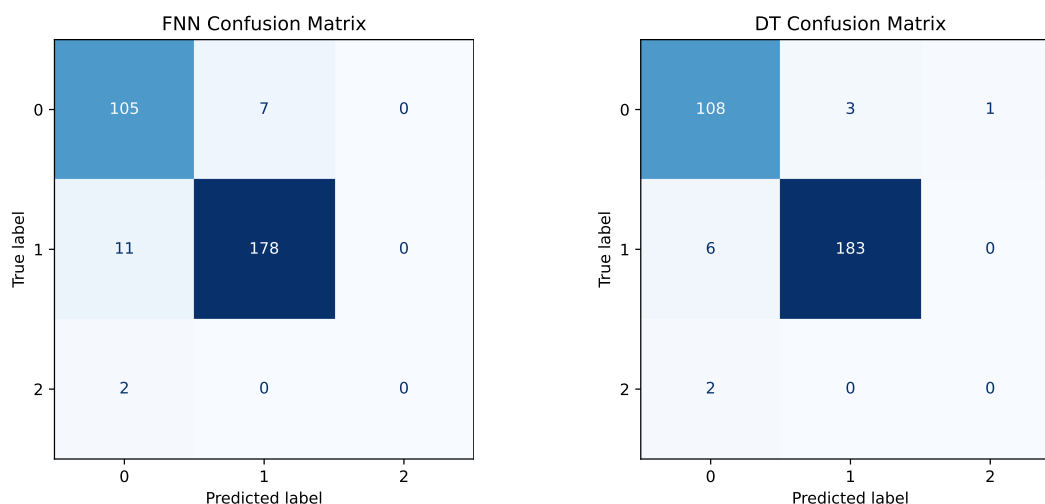


Table 4: Classwise metrics extracted from FNN's and DT's confusion matrices.

Model	0 (PD)			1 (Healthy)			2 (Prodromal)		
	Acc.*	Recall	F1-Score	Acc.*	Recall	F1-Score	Acc.*	Recall	F1-Score
FNN	84.00%	93.75%	91.30%	90.82%	94.18%	95.19%	0.00%	0.00%	0.00%
DT	90.00%	96.43%	94.74%	95.31%	96.83%	97.60%	0.00%	0.00%	0.00%

* Accuracy.

Optuna's optimization results were not discussed here due to being somewhat off-topic. For a discussion on Optuna's optimization and its resulting hyperparameters, check section 4.

► Question 3

Perform an experiment testing different kernel functions and hyperparameter configurations to obtain the best Support Vector Machine (SVM) classifier for a binary classification task, using a dataset of your choice. Don't forget to discuss the results.

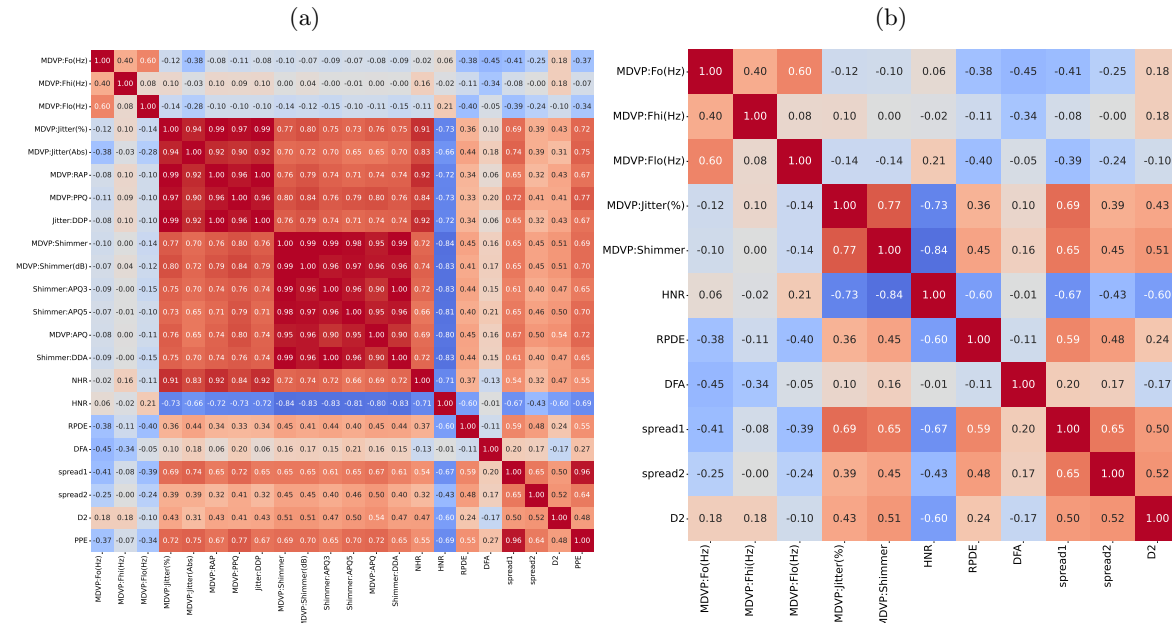
Data description. The dataset chosen is the Oxford Parkinson's Disease Detection Dataset, openly available at <https://archive.ics.uci.edu/dataset/174/parkinsons> (loaded in `parkinsons.csv`). It “is composed of a range of biomedical voice measurements from 31 people, 23 with Parkinson's disease (PD). Each column in the table is a particular voice measure, and each row corresponds one of 195 voice recording from these individuals ('name' column). The main aim of the data is to discriminate healthy people from those with PD, according to 'status' column which is set to 0 for healthy and 1 for PD”, as quoted directly from the authors.

The relatively low number of samples and high number of features play into the strengths of an SVM classifier. As done in the previous questions (see fig. 1), non-relevant features were dropped (e.g., the name column), highly correlated features were removed (see fig. 6), and the remaining samples shuffled before being the train-validation-test split. Table 5 assesses the size and composition of the resulting splits.

Table 5: Parkinson Detection dataset splits. Randomizer seed was fixed at 2421.

Class	Absolute and relative amounts within split					
	Train		Validation		Test	
0 (Healthy)	27	27.84%	9	18.75%	12	24.00%
1 (PD)	70	72.16%	39	81.25%	38	76.00%
Samples	97	49.74%	48	24.62%	50	25.64%

Figure 6: Feature correlations found in a subset of Parkinson Detection dataset (a) before and (b) after removing highly-correlated features.



As done in the previous questions, the Optuna framework was used to find the best hyperparameters with respect to overall accuracy on the validation set. The search space and best hyperparameters after 100 trials are shown in table 6; the associated model managed to obtain an error rate of 0.00%. The confusion matrix obtained on the test set from this model is shown in fig. 7. Table 7 lists some metrics derived from the results in the matrix.

Table 6: SVM hyperparameters.

Hyperparameter	Search space	Best after 100 trials
Kernel	Linear, polynomial, RBF* or sigmoid	RBF
Regularization constant**	$[10^{-1}, 10^3]$	17.72
Degree***	$\{2, 3, 4, 5\}$	-
Linear coefficient***	$[0, 10]$	-

* Radial Basis Function

** Sampled from the log domain

*** Valid only for the polynomial kernel

It is clear that the SVM was a good predictor of PD, but its performance on healthy patients is lacking. As seen previously, this is likely due to the use of an objective function in the validation stage that favors classes with greater number of samples.

Changing the objective function to consider each class separately but simultaneously should prove an improvement to this explorative study.

Figure 7: Confusion matrix of best SVM.

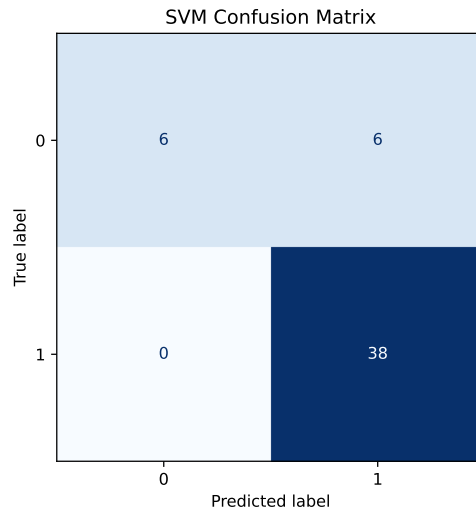


Table 7: Classwise metrics extracted from SVM's confusion matrix.

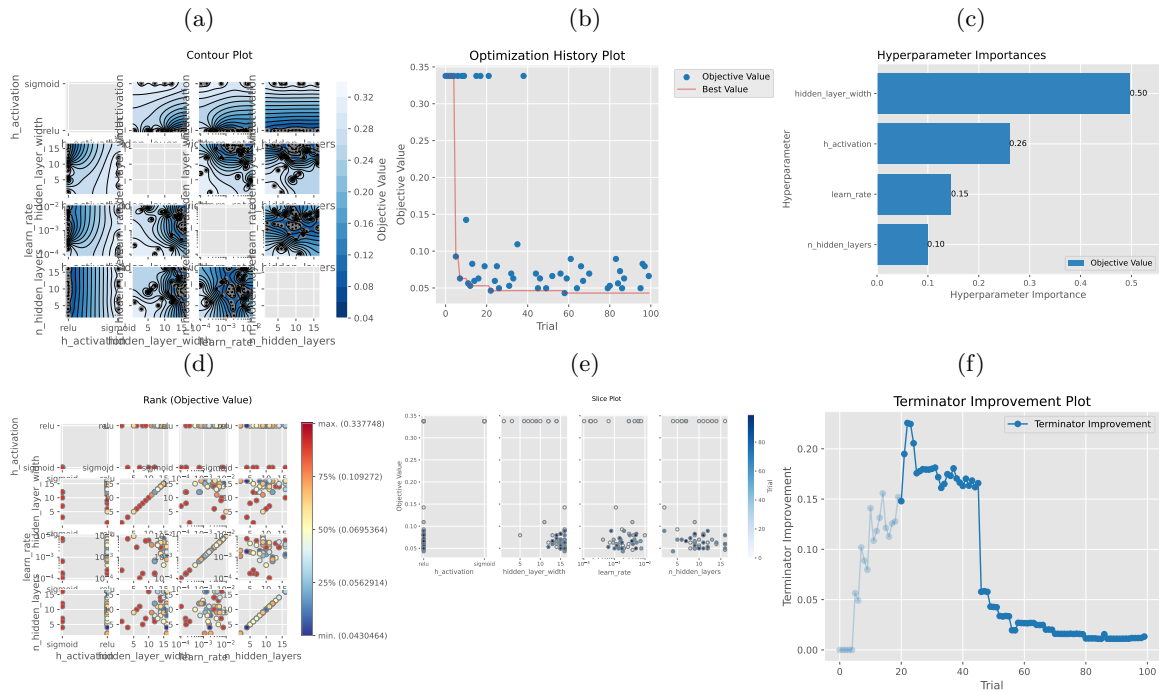
Class	Accuracy	Recall	Precision	F1-Score
0 (Healthy)	50.00%	50.00%	100.00%	66.67%
1 (PD)	86.36%	100.00%	86.36%	92.68%

Optuna's optimization results for this classifier included commentary on the importance of the kernel. To read about it, see section 4.

► **Optuna results**

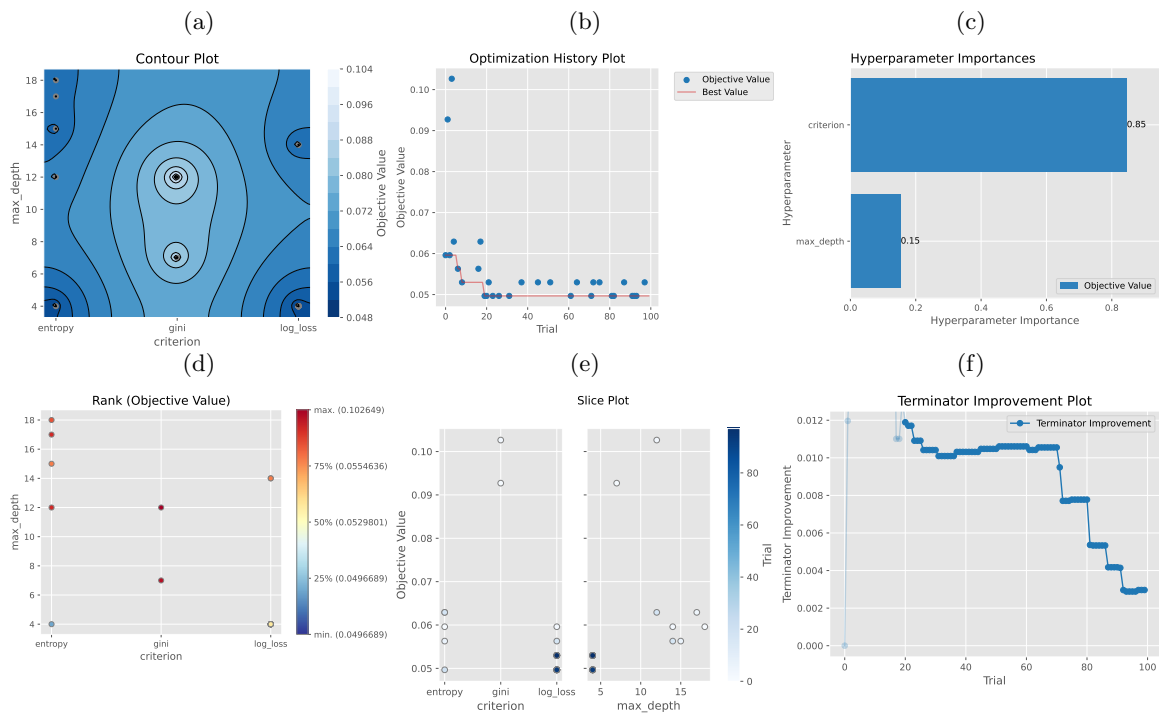
▷ **Feedforward Neural Network**

Figure 8: Optuna exploration on FNN hyperparameters. Objective value corresponds to error rate on validation set across all classes.



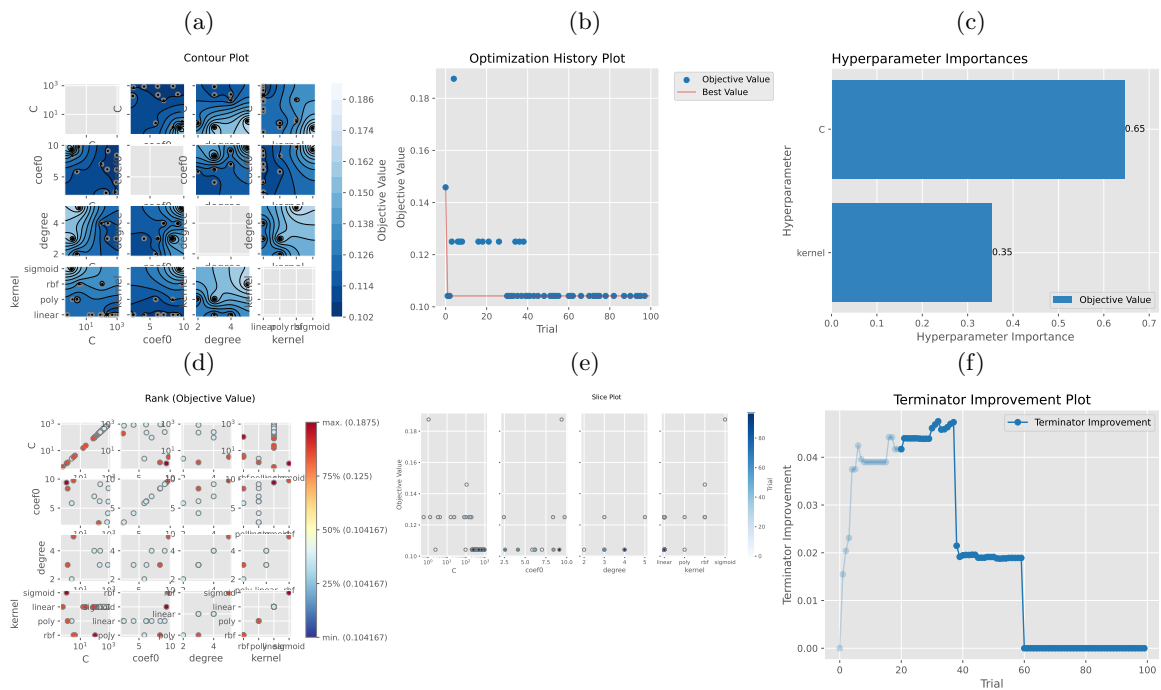
▷ Decision Tree

Figure 9: Optuna exploration on DT hyperparameters. Objective value corresponds to error rate on validation set across all classes.



▷ Support Vector Machine

Figure 10: Optuna exploration on SVM hyperparameters. Objective value corresponds to error rate on validation set across all classes.



► **Note on data**

▷ **PPMI features**